FEDERAL BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, ISLAMABAD

Annual Examination 2026

Chemistry — Class IX

Syllabus: Ch 1 — Nature of Chemistry | Ch 2 — Matter | Ch 3 — Atomic Structure | Ch 4 — Periodic Table & Periodicity | Ch 5 — Chemical Bonding | Ch 6 — Stoichiometry | Ch 7 — Electrochemistry | Ch 8 — Energetics | Ch 9 — Chemical Equilibrium

ANSWER KEY (Version 6)

Total Marks: 65 **Time Allowed: 2h 30m** **Version: 6**

SECTION A — Multiple Choice Questions Answer Key (12 × 1 = 12 Marks)

Q1. Branch that gauges pollutant behaviour and develops pollution-control methods: √ **B — Analytical Chemistry** **1 mark**

The textbook's own review question states that **Analytical Chemistry** "improves to gauge the behaviour of pollutants and develop techniques for pollution control." This is distinct from Environmental Chemistry, which focuses on protecting the environment (e.g. water purification) rather than measurement/analysis techniques.

Q2. Substance with DECREASING solubility as temperature rises: √ **C — SO_2** **1 mark**

On the solubility curves (Fig. 2.4), **gases such as SO_2** show a downward-sloping curve: their solubility falls as temperature rises, because heating drives dissolved gas molecules out of solution. KNO_3, NH_4Cl, and NaCl are solids whose curves rise (or stay roughly flat) with increasing temperature.

Q3. Neutrons in $^{27}{}_{13}M$: √ **B — 14** **1 mark**

Neutrons = mass number − atomic number = **27 − 13 = 14**. The atomic number (13, lower-left) gives protons; the mass number (27, upper-left) gives the total of protons + neutrons.

Q4. Statement NOT correct about isotopes: √ **D — Same physical properties** **1 mark**

Isotopes share the **same atomic number, same number of protons, and the same chemical properties** (since chemical behaviour depends on electron configuration). However, they have **different numbers of neutrons and different mass numbers**, which gives them different **physical properties** such as density and, in some cases, radioactivity.

Q5. Valence electrons of $^{32}{}_{16}S$: √ **C — 6** **1 mark**

Sulphur (Z=16) has electronic configuration 2,8,6. For normal (main-group) elements, **valence electrons = group number**, so sulphur has **6 valence electrons** and belongs to Group 16 (VIA).

Q6. Greatest electron affinity among S, P, Cl: √ **C — Cl** **1 mark**

Electron affinity generally **increases across a period** (left to right) because of increasing nuclear charge and decreasing atomic size, which pulls an incoming electron in more strongly. Since S, P, and Cl are all Period 3 elements and Cl is furthest to the right, **chlorine** has the greatest electron affinity of the three.

Q7. Valence electrons of silicon (Group IVA): √ **C — 4** **1 mark**

Silicon belongs to **Group IVA**, and for normal elements the group number (in the old A/B notation, the "4" in IVA) gives the number of valence electrons: silicon has **4 electrons** in its valence shell ($3s^2 3p^2$).

Q8. Atom that will NOT form a cation or anion: √ **C — Atomic No. 18** **1 mark**

Atomic number 18 is **argon**, a noble gas with a complete valence shell (2,8,8). Having a full octet already, it has no tendency to lose or gain electrons and does **not** form ions. Atoms 16 (S), 17 (Cl), and 19 (K) all have incomplete valence shells and readily form ions.

Q9. Mass of carbon in 44 g CO_2: √ **C — 12 g** **1 mark**

Molar mass of CO_2 = 12 + 2(16) = **44 g/mol**, so 44 g CO_2 = **1 mole**. Each mole of CO_2 contains 1 mole of carbon atoms, whose mass = **12 g** (the gram atomic mass of carbon).

Q10. Term the same for 1 mole O_2 and 1 mole H_2O: √ **C — Number of molecules** **1 mark**

By definition, **one mole of any substance contains 6.022×10^{23} representative particles** (Avogadro's number). Since both O_2 and H_2O exist as molecules, one mole of each contains exactly **6.022×10^{23} molecules** — even though their masses, volumes, and atom counts differ.

Q11. Acids formed from SO_2 and NO_2 in acid rain: √ **B — H_2SO_4 & HNO_3** **1 mark**

Air pollutants SO_2 and NO_2 combine with oxygen and water vapour in the atmosphere to form **sulphuric acid (H_2SO_4)** and **nitric acid (HNO_3)** respectively. These acids fall to the ground with rain, making it acidic, and can travel hundreds of kilometres from the source of pollution.

Q12. The "system" in the limestone + HCl example: √ **B — Limestone and HCl solution** **1 mark**

In chemistry, a **system** is the specific part of the universe under study — here, the **limestone and hydrochloric acid solution** reacting inside the test tube. The **test tube and everything around it** (including the room and heat source) form the **surroundings/environment**, not the system itself.

SECTION B — Short Questions Answer Key (11 × 3 = 33 Marks)

2(i) Forensic chemistry / Electrochemistry as physical chemistry **3 marks**

Forensic chemistry helps solve crimes by **analysing physical evidence** (such as blood, fibres, or trace chemicals) found at a crime scene to identify substances and link them to suspects or events. **2 marks**
It is a specific application of the broader branch **Analytical Chemistry**. **1 mark**

OR

Vehicle batteries work on the principle of **electrochemistry**, which is an application of **Physical Chemistry** (the branch dealing with the laws and theories that explain the structure and changes of matter). **2 marks**
Another daily-life application of physical chemistry: understanding **reaction rates** (e.g. why food spoils faster in warm weather) or **energy changes** in reactions (any reasonable example accepted). **1 mark**

2(ii) Homogeneous vs. heterogeneous solution / Solution as a mixture **3 marks**

A **homogeneous solution** has uniform composition throughout, with solute particles evenly distributed at the molecular level, e.g. **salt dissolved in water**. **1.5 marks**
A **heterogeneous solution/mixture** does not have uniform composition; different parts can be visibly distinguished, e.g. **sand mixed with iron filings**. **1.5 marks**

OR

A solution is considered a **mixture** because it consists of two or more substances (solute + solvent) that are **physically combined** in a non-fixed ratio and can be separated by physical means (e.g. evaporation) — unlike a compound, where components are chemically bonded in a fixed ratio. **1.5 marks**
Test: Shine a beam of light through the liquid. If the light beam is visible (scattered) as it passes through, the liquid is a **colloid** (Tyndall effect). If the light passes through without scattering, it is a **true solution**. **1.5 marks**

2(iii) Shell vs. sub-shell / M-shell notation 3 marks

A **shell** (principal energy level, n = 1,2,3...) is a region around the nucleus where electrons are likely to be found; a **sub-shell** is a division within a shell (s, p, d, f), each holding electrons of slightly different energy. 1.5 marks

The **N-shell (n=4)** contains **4 sub-shells: 4s, 4p, 4d, 4f**. 1.5 marks

OR

The **M-shell (n=3)** contains the sub-shells **3s, 3p, 3d**. 1.5 marks

In order of **increasing energy**: **3s < 3p < 3d**. 1.5 marks

2(iv) p sub-shell capacity + Na configuration / Identifying an atom without neutrons 3 marks

The **p sub-shell** can accommodate a maximum of **6 electrons**. 1 mark

Electronic configuration of sodium (Z=11): $\mathbf{1s^2 2s^2 2p^6 3s^1}$. 2 marks

OR

Yes, an atom can be identified without knowing its number of neutrons. 1 mark

The identity of an atom (which element it is) is determined by its **atomic number**, which equals the number of **protons** in its nucleus (or equivalently, the number of electrons in a neutral atom). The number of neutrons only distinguishes between different **isotopes** of the same element — it does not change which element the atom is. 2 marks

2(v) Shielding effect (Li vs Na) / Ionization energy (Na vs K) 3 marks

Sodium has a higher shielding effect than lithium. 1 mark

Sodium (2,8,1) has more inner electron shells (2 complete shells) between its nucleus and valence electron than lithium (2,1), which has only 1 inner shell. More inner shells means **greater shielding** of the nuclear charge felt by the valence electron. 2 marks

OR

Sodium has a **higher ionization energy** than potassium. 1 mark

Down Group 1, from Na to K, an extra electron shell is added, increasing atomic size and shielding effect. This means potassium's valence electron is **farther from the nucleus and more shielded**, so it is held less tightly and removed more easily (lower IE) than sodium's valence electron. 2 marks

2(vi) Group number from configuration / General valence configurations 3 marks

Aluminium ($^{27}_{13}Al$): configuration 2,8,3 → 3 valence electrons → **Group 13 (IIIA)**. 1.5 marks

Potassium ($^{39}_{19}K$): configuration 2,8,8,1 → 1 valence electron → **Group 1 (IA)**. 1.5 marks

OR

- **Alkali metals:** ns^1 ¾ mark
- **Alkaline earth metals:** ns^2 ¾ mark
- **Halogens:** ns^2np^5 ¾ mark
- **Noble gases:** ns^2np^6 (except He, which is $1s^2$) ¾ mark

2(vii) Octet/duplet rule (H exception) / Phosphorus octet **3 marks**

Octet rule: atoms tend to gain, lose, or share electrons to achieve **8 electrons** in their valence shell (the configuration of the nearest noble gas). **1 mark**

Duplet rule: very small atoms (with only the K-shell available) achieve stability with just **2 electrons** instead of 8. **0.5 mark**

Among O, F, H, and N, **hydrogen (H)** obeys the duplet rule — because it has only 1 shell (K-shell), which is complete with just 2 electrons (like helium), so it cannot hold 8. **1.5 marks**

OR

Phosphorus (Group VA, Period 3) has **5 valence electrons** and needs **3 more electrons** to complete its octet (5 + 3 = 8). **1.5 marks**

Electron-dot structure: P with 5 dots arranged around the symbol (one dot on each of 4 sides, with one side carrying a pair): **P** surrounded by 5 unpaired/paired valence electrons (2 paired + 3 single dots). **1.5 marks**

2(viii) Na_2S formation (identify false statement) / O and Mg ions **3 marks**

The **incorrect (NOT true) statement is (d)**: "each sulphur atom gains one electron." **1 mark**

Correct formation: Sulphur (2,8,6) needs **2 electrons** to complete its octet, not one. In Na_2S, **two sodium atoms** each lose 1 electron (2,8,1 → 2,8, forming Na^+), and these **2 electrons together** are gained by **one sulphur atom** (2,8,6 → 2,8,8, forming S^{2-}). The compound formula is Na_2S. **2 marks**

OR

Atomic number 8 (oxygen): configuration 2,6 → gains 2 electrons to complete its octet → forms **O^{2-}** ion (charge −2), configuration becomes 2,8. **1.5 marks**

Atomic number 12 (magnesium): configuration 2,8,2 → loses 2 electrons to attain a stable octet → forms **Mg^{2+}** ion (charge +2), configuration becomes 2,8. **1.5 marks**

2(ix) Molecules in benzene / Molecules in acetic acid **3 marks**

Number of molecules = moles × N_A = 1.09 × 6.022×10^{23} **1.5 marks**

= **6.564×10^{23} molecules** of benzene **1.5 marks**

OR

Number of molecules = 0.01 × 6.022×10^{23} **1.5 marks**

= **6.022×10^{21} molecules** of acetic acid **1.5 marks**

2(x) Mass of $CuSO_4 \cdot 5H_2O$ / Molecules in steam **3 marks**

Molar mass of $CuSO_4 \cdot 5H_2O$ = 63.5 + 32 + 4(16) + 5(18) = 63.5 + 32 + 64 + 90 = **249.5 g/mol** **1.5 marks**

Mass = moles × molar mass = 1.05 × 249.5 = **261.975 g ≈ 262.0 g** **1.5 marks**

OR

Molar mass of H_2O = 2(1) + 16 = 18 g/mol. Moles = 9.0 / 18 = **0.5 mol** **1.5 marks**

Molecules = 0.5 × 6.022×10^{23} = **3.011×10^{23} molecules** **1.5 marks**

2(xi) Acid rain formation / Oxidation state of N in $NaNO_2$ **3 marks**

Air pollutants **SO_2 and NO_2** combine with **oxygen and water vapour** in the atmosphere. **1 mark**

These reactions form **H_2SO_4** (from SO_2) and **HNO_3** (from NO_2). **1 mark**

These acids fall to the ground with rain (sometimes carried hundreds of kilometres by clouds first), making the rain acidic — this is called **acid rain**. **1 mark**

OR

Let oxidation state of N = x. Na is +1, O is −2 (×2 atoms). **1 mark**

(+1) + x + 2(−2) = 0 → 1 + x − 4 = 0 → x − 3 = 0 **1 mark**

x = +3. The oxidation state of nitrogen in $NaNO_2$ is **+3**. **1 mark**

SECTION C — Long Questions Answer Key (4 × 5 = 20 Marks)

3(i) Protons/neutrons/electrons of P / N-shell sub-shells / Ca atom classification **5 marks**

(a) Z=15, A=31: [2 marks]

Protons = atomic number = **15**. **⅔ mark**
Neutrons = mass number − atomic number = 31 − 15 = **16**. **⅔ mark**
Electrons = protons (neutral atom) = **15**. **⅔ mark**

(b) N-shell sub-shells: [3 marks]

The N-shell (n=4) contains **4 sub-shells: 4s, 4p, 4d, 4f**. **2 marks**
The **f sub-shell** can hold a maximum of **14 electrons**. **1 mark**

OR

(a) 20 protons, mass number 40: [2 marks]

Neutrons = 40 − 20 = **20**. **1 mark**
Electrons = 20 (neutral atom). **1 mark**

(b) Configuration, group/period, metal/non-metal: [3 marks]

This is **calcium**. Electronic configuration: **$1s^2 2s^2 2p^6 3s^2 3p^6 4s^2$**. **1.5 marks**
2 valence electrons → **Group 2**; highest shell n=4 → **Period 4**. It is a **metal** (alkaline earth metal), since it has few valence electrons and readily loses them to form Ca^{2+}. **1.5 marks**

3(ii) Pairing elements by similar chemical properties / Ionization energy matching **5 marks**

(a) Four pairs with similar chemical properties: [3 marks]

Pair	Elements	Group
A & H	A = $1s^2 2s^2$ (Be), H = $1s^2 2s^2 2p^6 3s^2$ (Mg)	Group 2 (alkaline earth metals)
B & D	B = $1s^2 2s^2 2p^6$ (Ne), D = $1s^2$ (He)	Group 18 (noble gases)
C & E	C = $1s^2 2s^2 2p^3$ (N), E = $1s^2 2s^2 2p^6 3s^2 3p^3$ (P)	Group 15 (VA)
F & G	F = $1s^2 2s^1$ (Li), G = $1s^2 2s^2 2p^6 3s^1$ (Na)	Group 1 (alkali metals)

¾ mark per correctly matched and named pair

(b) Classification: [2 marks]

A & H (Be, Mg): **metals**. F & G (Li, Na): **metals**. C & E (N, P): **non-metals**. B & D (Ne, He): **noble gases**. **0.5 mark each**

OR

(a) Matching ionization energies: [3 marks]

$1s^2 2s^2 2p^4$ (oxygen, Z=8) → 2080 kJ/mole (the higher value). **1.5 marks**
$1s^2 2s^2 2p^6 3s^1$ (sodium, Z=11) → 496 kJ/mole (the lower value). **1.5 marks**

(b) Reasoning: [2 marks]

Oxygen's valence electron is in the **2p sub-shell, close to the nucleus with little shielding**, so it experiences a strong effective nuclear pull and requires much more energy to remove — giving it the **higher** ionization energy. **1 mark**
Sodium's valence electron is in the **3s sub-shell, an extra shell further from the nucleus and shielded** by the full 2,8 core, so it is held much more loosely and removed with far less energy — giving it the **lower** ionization energy. **1 mark**

3(iii) Formation of AlF_3 / Cation formation for Sr and Ba **5 marks**

(a) Formation of AlF_3: [3 marks]

Aluminium (Z=13): 2,8,3 has 3 valence electrons. Fluorine (Z=9): 2,7 has 7 valence electrons (needs 1 more). **1 mark**

Each Al atom **loses 3 electrons** (one to each of three F atoms) to form Al^{3+} (configuration 2,8). Each of the **three F atoms gains 1 electron** to form F^- (configuration 2,8). **1.5 marks**

Overall: $Al \rightarrow Al^{3+} + 3e^-$; $3F + 3e^- \rightarrow 3F^-$, combining to give AlF_3 (dot-and-cross diagram: Al^{3+} with no valence dots, surrounded by three F^- ions each showing 8 dots/crosses). **0.5 mark**

(b) Anion from atomic number 16 (sulphur): [2 marks]

Sulphur (2,8,6) gains **2 electrons** to complete its octet, forming an ion of charge **−2 (S^{2-})**. **1 mark**

Electron configuration of S^{2-}: **2,8,8**. **1 mark**

OR

(a) Cation formation for Sr and Ba: [3 marks]

Strontium (Z=38, Group 2) has 2 valence electrons and loses both to form Sr^{2+} (stable noble-gas-like configuration). **1.5 marks**

Barium (Z=56, Group 2) similarly has 2 valence electrons and loses both to form Ba^{2+}. **1.5 marks**

(Electron-dot diagram: Sr and Ba shown with 2 dots initially, both removed, leaving $[Sr]^{2+}$ and $[Ba]^{2+}$ with no valence dots.)

(b) Silicon: ionic or covalent? [2 marks]

Silicon is more likely to form **covalent bonds**. **1 mark**

With 4 valence electrons, silicon lies in the middle of Period 3 and is a **metalloid**; it would need to lose or gain 4 electrons to reach a stable octet, which requires too much energy. Instead, it **shares** electrons with other non-metals (as in SiO_2), forming covalent bonds. **1 mark**

3(iv) Molar mass and empirical formula of indigo/indoxyl / Phosphorus oxide **5 marks**

(a) Molar mass of indigo, $C_{16}H_{10}N_2O_2$: [2 marks]

= 16(12) + 10(1) + 2(14) + 2(16) = 192 + 10 + 28 + 32 **1 mark**

= 262 g/mol **1 mark**

(b) Molar mass of indoxyl, C_8H_7ON: [1 mark]

= 8(12) + 7(1) + 16 + 14 = 96 + 7 + 16 + 14 = **133 g/mol** **1 mark**

(c) Empirical formulas: [2 marks]

Indigo $C_{16}H_{10}N_2O_2$: HCF of 16,10,2,2 is 2 → divide throughout → C_8H_5NO. **1 mark**

Indoxyl C_8H_7ON: HCF of 8,7,1,1 is 1 → already simplest ratio → empirical formula = C_8H_7NO (same as molecular formula). **1 mark**

OR

(a) Molecular and empirical formula: [2 marks]

Molecular formula: P_4O_{10} (4 P atoms, 10 O atoms). **1 mark**

HCF of 4 and 10 is 2 → empirical formula = P_2O_5. **1 mark**

(b) Molar mass: [2 marks]

= 4(31) + 10(16) = 124 + 160 **1 mark**

= 284 g/mol **1 mark**

(c) Molecular formulas from structural formulas: [1 mark]

$CH_3–CH_2–OH$ → C_2H_6O (ethanol); $CH_3–CO–CH_3$ → C_3H_6O (acetone). **0.5 mark each**